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1KCQ | pdb_00001kcq

Human Gelsolin Domain 2 with a Cd2+ bound

PDB DOI: https://doi.org/10.2210/pdb1KCQ/pdb

Classification: STRUCTURAL PROTEIN
Organism(s): Homo sapiens
Expression System: Escherichia coli
Mutation(s): No

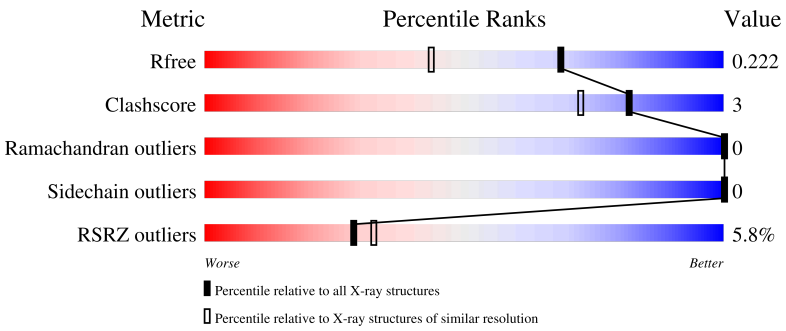
Deposited: 2001-11-09 **Released:** 2002-01-04
Deposition Author(s): Kazmirski, S.L., Isaacson, R.L., An, C., Buckle, A., Johnson, C.M., Daggett, V., Fersht, A.R.

Experimental Data Snapshot

Method: X-RAY DIFFRACTION
Resolution: 1.65 Å
R-Value Free:
0.233 (Depositor), 0.222 (DCC)
R-Value Work:
0.177 (Depositor), 0.171 (DCC)
R-Value Observed:
0.180 (Depositor)

Starting Model: experimental
View more details

wwPDB Validation 3D Report | Full Report



This is version 1.4 of the entry. See complete history.

Literature

Loss of a metal-binding site in gelsolin leads to familial amyloidosis-Finnish type.

Kazmirski, S.L., Isaacson, R.L., An, C., Buckle, A., Johnson, C.M., Daggett, V., Fersht, A.R.
(2002) Nat Struct Biol **9**: 112-116

PubMed: 11753432
DOI: https://doi.org/10.1038/nsb745
Primary Citation Related Structures:
1KCQ

PubMed Abstract:
Mutations in domain 2 (D2, residues 151-266) of the actin-binding protein gelsolin cause familial amyloidosis-Finnish type (FAF). These mutations, D187N or D187Y, lead to abnormal proteolysis of plasma gelsolin at residues 172-173 and a second hydrolysis at residue 243, resulting in an amyloidogenic fragment. Here we present the structure of human gelsolin D2 at 1.65 A and find that Asp 187 is part of a Cd2+ ...
View More

Organizational Affiliation:
Cambridge University Chemical Laboratory and Cambridge Centre for Protein Engineering, MRC Centre, Hills Road, Cambridge CB2 2QH, UK.

Biological Assembly 1



Explore in 3D: Structure | Sequence Annotations | Electron Density | Validation Report | Ligand Interaction (CD)

Global Symmetry: Asymmetric - C1
Global Stoichiometry: Monomer - A1

Find Similar Assemblies

Biological assembly 1 assigned by authors.

Macromolecule Content

- Total Structure Weight: 12.04 kDa
- Atom Count: 937
- Modeled Residue Count: 104
- Deposited Residue Count: 104
- Unique protein chains: 1

Macromolecules

Find similar proteins by: Sequence | 3D Structure

Entity ID: 1

Molecule	Chains	Sequence Length	Organism	Details
GELSOLIN	A	104	Homo sapiens	**Mutation(s)**: 0

UniProt & NIH Common Fund Data Resources

Find proteins for P06396 *(Homo sapiens)* Explore P06396

PHAROS: P06396 GTEx: ENSG00000148180

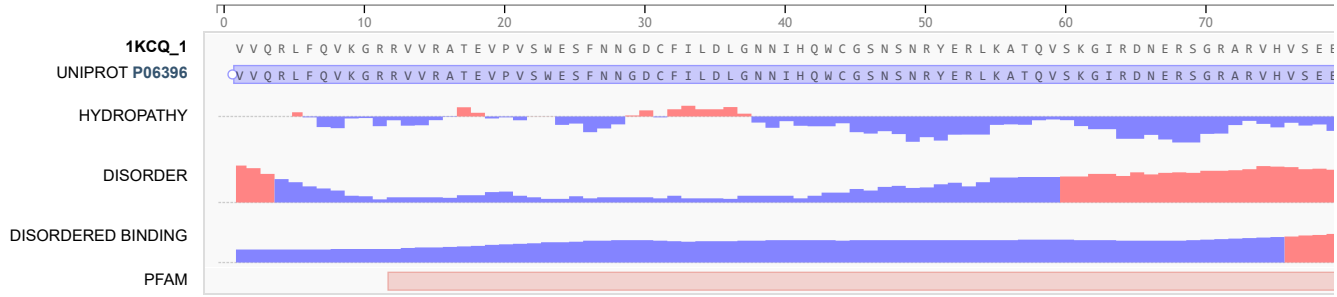
Entity Groups

Sequence Clusters: 30% Identity | 50% Identity | 70% Identity | 90% Identity | 95% Identity | 100% Identity

UniProt Group	P06396

Sequence Annotations

Reference Sequence 1KCQ_1



Small Molecules

Ligands 1 Unique

ID	Chains	Name / Formula / InChI Key	2D Diagram	3D Interactions
CD Download: Instance Coordinates	B [auth A], C [auth A], D [auth A], E [auth A]	**CADMIUM ION** Cd WLZRMCYVCSSEQC-UHFFFAOYSA-N	Cd^{+2}	Interactions Interactions & Density

Experimental Data & Validation

Experimental Data

Method: X-RAY DIFFRACTION
Resolution: 1.65 Å
R-Value Free: 0.233 (Depositor), 0.222 (DCC)
R-Value Work: 0.177 (Depositor), 0.171 (DCC)
R-Value Observed: 0.180 (Depositor)

Space Group: C 1 2 1

Unit Cell:

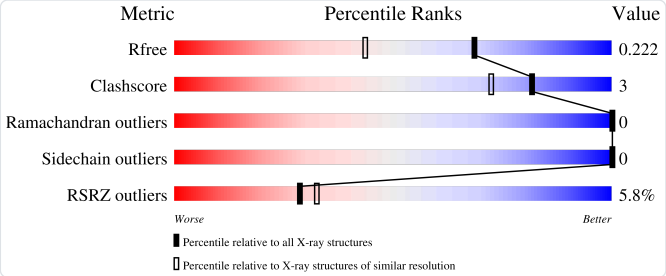
Length (Å)	Angle (°)
a = 96.968	α = 90
b = 26.525	β = 121.19
c = 50.294	γ = 90

Software Package:

Software Name	Purpose
MAR345	data collection
SCALA	data scaling
AMoRE	phasing
X-PLOR	refinement
CCP4	data scaling

Structure Validation

View Full Validation Report



Entry History

Deposition Data

Released Date: 2002-01-04
Deposition Author(s): Kazmirski, S.L., Isaacson, R.L., An, C., Buckle, A., Johnson, C.M., Daggett, V., Fersht, A.R.

Revision History (Full details and data files)

- **Version 1.0: 2002-01-04**
  Type: Initial release
- **Version 1.1: 2008-04-27**
  Changes: Version format compliance
- **Version 1.2: 2011-07-13**
  Changes: Derived calculations, Version format compliance
- **Version 1.3: 2023-08-16**
  Changes: Data collection, Database references, Derived calculations, Refinement description
- **Version 1.4: 2024-11-20**
  Changes: Structure summary

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Partnerships & Collaborators:

RCSB PDB Core Operations are funded by the U.S. National Science Foundation (DBI-2321666), the US Department of Energy (DE-SC0019749), and the National Cancer Institute, National Institute of Allergy and Infectious Diseases, and National Institute of General Medical Sciences of the National Institutes of Health under grant R01GM157729. RCSB PDB uses resources of the National Energy Research Scientific Computing Center (NERSC), a Department of Energy User Facility.